

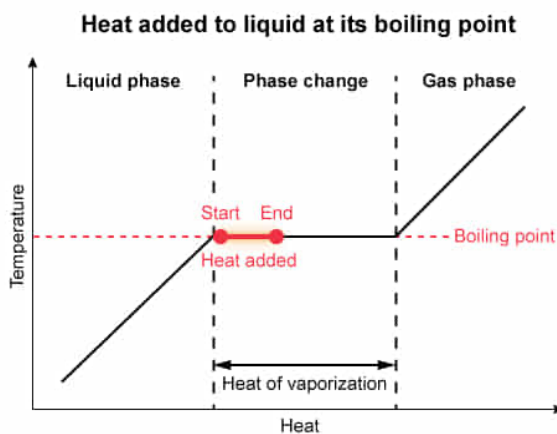
If a liquid is at its exact boiling point, what will happen to the liquid when a small amount of heat is added to it?

- ☐ A. None of the liquid will turn to gas, and the temperature will increase. [3%]
- ☐ B. All the liquid will turn to gas, and the temperature will remain the same. [10%]
- ☐ C. A small amount of the liquid will turn to gas, and the temperature will increase. [26%]
- ✔ ☒ D. A small amount of the liquid will turn to gas, and the temperature will remain the same. [59%]

Correct

60% answered correctly

Explanation:



Adding a small amount of heat to a liquid at its boiling point:

Small amount of liquid turns to gas
Temperature remains the same

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Matter typically exists in one of three phases: solid, liquid, or gas. Matter transitioning from one phase to another undergoes a **phase change** in which it absorbs or releases **heat** until all the matter completes the transition to the new phase. During a phase change, the **temperature** of the material remains **constant** even though heat is added or removed. Matter transitions from a liquid to a gas at its boiling point and requires the addition of heat energy equal to its **heat of vaporization** for all the liquid to transition into gas.

In this question, the temperature of the liquid equals its exact boiling point. As a result, adding a small amount of heat will cause some of the liquid to transition into gas, but the temperature will not change.

(Choices A and C) The temperature will remain constant until the heat of vaporization is exceeded and all the liquid has turned to gas.

(Choice B) A small amount of heat will convert a small amount of the liquid to gas, not all the liquid.

Educational objective:

During phase transitions, the temperature of a substance remains constant. A liquid at the exact temperature of its boiling point is transitioning from a liquid to a gas. It must gain an amount of heat equal to its heat of vaporization before all the liquid turns to gas and the temperature increases.

Time spent:0 sec

Subject:

Physics

QID:404155

System:

Thermodynamics